Identification of a Feasible, Evidence-Based Approach to Ulcerative Colitis Disease Assessment by Specialty Pharmacies

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Introduction

- Ulcerative colitis (UC) is an inflammatory bowel disease (IBD) of the colon characterized by bloody diarrhea and abdominal pain[1]
- Treat-to-target recommendations provide a clinical framework to assess treatment goals by using patient-reported outcomes (PROs) and objective markers to monitor therapeutic response, guide treatment decisions, and promote shared decision making[2–4]
- Specialty pharmacies play a crucial role in managing UC by assessing disease activity through PROs that may be used to inform short-, intermediate-, and long-term treatment targets
- A standardized approach to UC disease assessment has not been widely adopted in clinical practice

Objective

- This project aimed to identify and align on evidence-based clinical parameters for use by specialty pharmacies when assessing patients with UC, and to evaluate their real-world application

Methods

- The project comprised three sequential phases (**Figure 1**)

Figure 1. Project methodology overview

1 Targeted Literature Review

PubMed review of **published** evidence on **UC disease assessment** within the last 5 years using medical subject headings (MeSH) and keyword search terms related to **UC, disease assessment, monitoring,** and **treatment outcomes**

2 Evidence Council

Specialty pharmacy clinical application insights on the Targeted Literature Review findings from **five specialty pharmacy and/or IBD subject matter experts** over a 2-hour virtual session

3 Database Prevalence Analysis

Assessment of the **availability of Evidence Council recommended parameters** for specialty pharmacy assessment in Loopback's Therapy-Specific Outcomes Coalition (TSOC) **de-identified healthcare database** of 16 health systems

Results

Targeted Literature Review

- Of 296 articles identified, 39 were included, comprising 18 clinical trials, 6 systematic reviews, 9 meta-analyses, and 6 review articles
- Mayo score and its variations were the most reported disease activity tools (n=33) to assess clinical response and remission
- Biomarkers, including fecal calprotectin and C-reactive protein, were each reported in 13 publications
- There was consensus in using Mayo endoscopic subscore to assess endoscopic improvement or healing (n=28)
- The IBD questionnaire was the most reported quality of life assessment (n=9)

Evidence Council

- The Evidence Council agreed that standardizing disease assessments and implementing treat-to-target strategies should be a priority for clinical practice sites; see **Figure 2** for a summary of the key takeaways

Figure 2. Summary of Evidence Council key takeaways

1. There is **significant variability** in **UC disease assessment tools** used in clinical practice, most notably for **disease activity, PROs,** and **quality of life**
2. Current **specialty pharmacy UC disease assessments** are mostly **subjective** and **symptom-based**
3. **Standardizing** the use of simple **UC disease assessment tools** across clinical practice sites could lead to more **consistent** and **objective patient assessments, improved treatment monitoring,** and **better collaboration** among providers
4. Real-world **barriers to standardization** include **provider education, patient responsiveness,** and **ease of implementation** into electronic health record systems
5. **Easy-to-use** platforms such as a **clinical dashboard** are recommended to **standardize tracking** of **disease activity** and to incorporate **treat-to-target strategies**
6. **Evidence Council** members **recommend incorporating** the following parameters into the specialty pharmacy **clinical dashboard** to facilitate assessment of **UC disease activity**:[a]
 - **Modified Mayo score, patient-reported outcomes 2 (PRO2),** and their **rectal bleeding** and **stool frequency** components
 - **Fecal calprotectin** and **C-reactive protein**
 - **Healthcare resource utilization** and **steroid use**

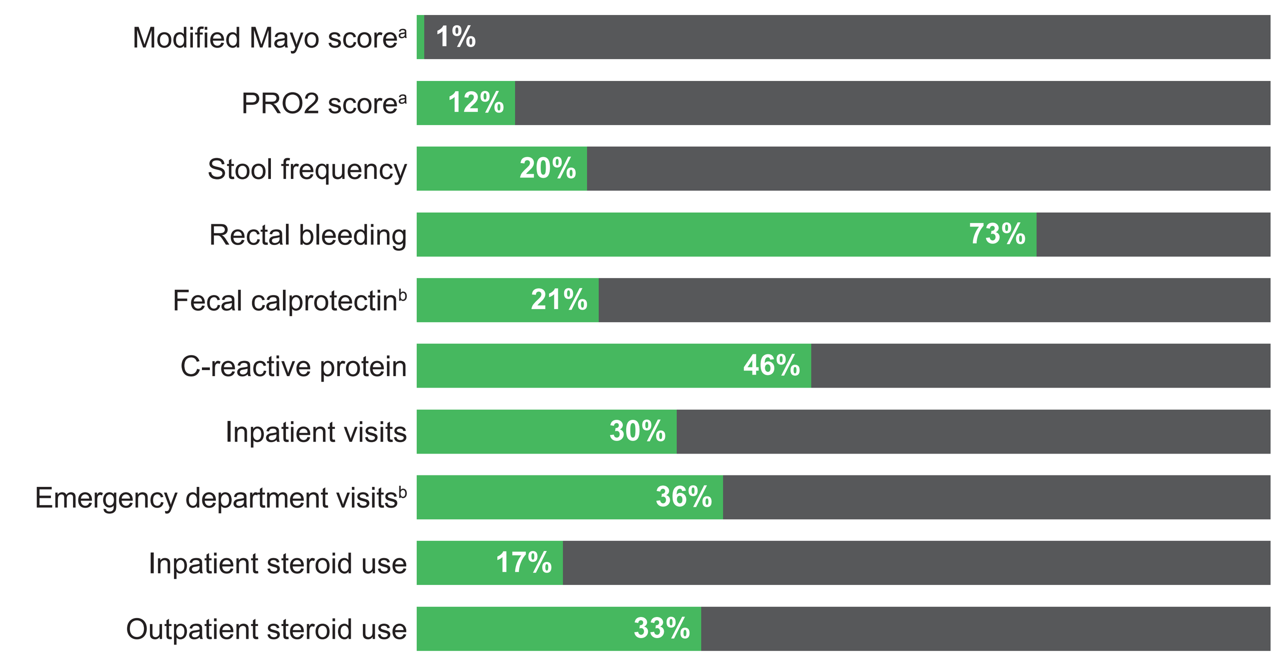
Evidence Council participants were chosen based on their IBD/gastroenterology clinical expertise and backgrounds through Loopback's network of health system specialty pharmacies.

[a]Assessment parameters were selected if ⩾3 Evidence Council members responded that the outcome should be included as part of the clinical dashboard to facilitate the assessment of disease activity and severity in patients with UC.

Database Prevalence Analysis

- A total of 62,097 patients with UC were identified in the database
- Among parameters recommended by the Evidence Council, prevalence was highest for rectal bleeding (73%) and lowest for modified Mayo score (1%) (**Figure 3**)
- Disease assessment reporting was inconsistent between the 16 health systems (**Figure 4**)

Figure 3. Prevalence of disease assessment parameters recommended by the Evidence Council among patients with UC in the TSOC database

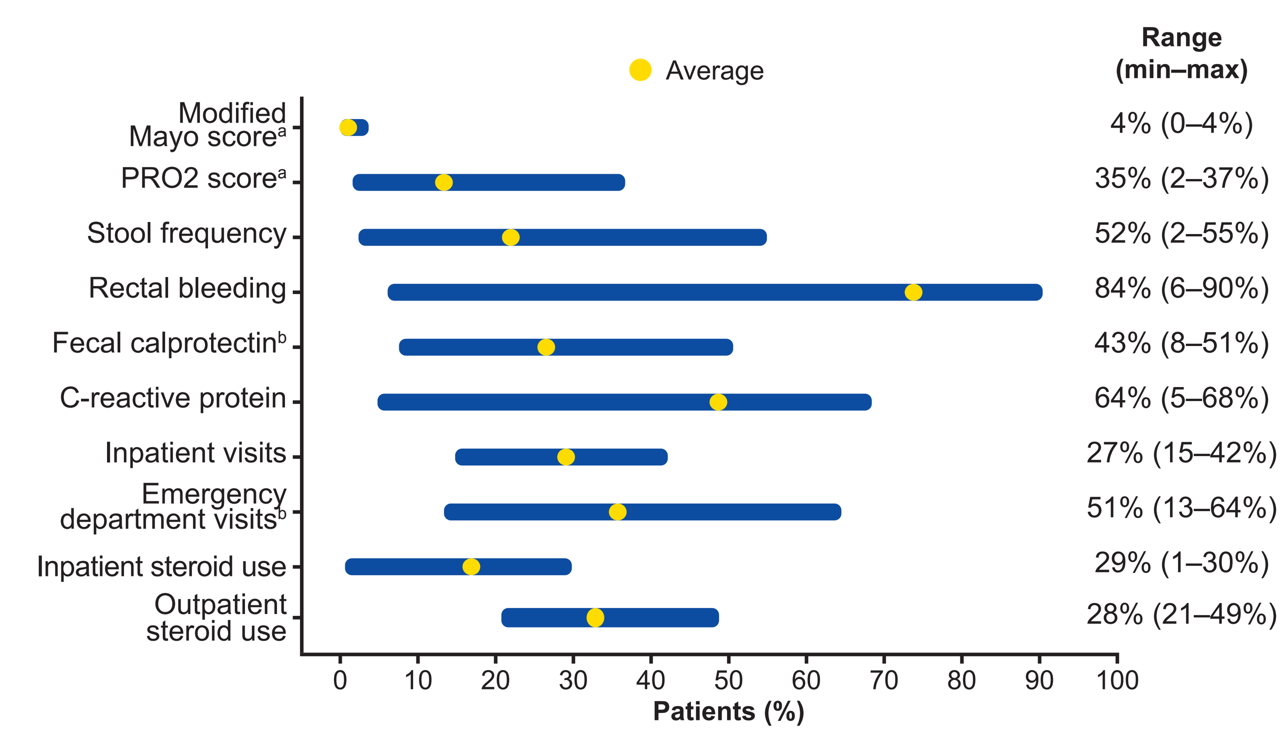

Modified Mayo score[a] 1%
PRO2 score[a] 12%
Stool frequency 20%
Rectal bleeding 73%
Fecal calprotectin[b] 21%
C-reactive protein 46%
Inpatient visits 30%
Emergency department visits[b] 36%
Inpatient steroid use 17%
Outpatient steroid use 33%

[a]Present or calculated.

[b]One client was excluded due to absence of data availability.

Loopback's TSOC data were extracted from electronic health records and pharmacy dispensing software, encompassing inpatient/outpatient clinical notes, laboratory results, and medication administrations, orders, and fills (as of January 3, 2025). Patients with UC were identified if they had any inpatient or outpatient encounter with a primary diagnosis code of K51. Disease assessments (modified Mayo score, PRO2, stool frequency, and rectal bleeding) were extracted from clinical notes using regular expressions for complex string-matching queries that searched for specific text patterns.

Figure 4. Patients with available UC disease assessment parameter data across 16 health systems in the TSOC database


Average
Range (min–max)
Modified Mayo score[a] 4% (0–4%)
PRO2 score[a] 35% (2–37%)
Stool frequency 52% (2–55%)
Rectal bleeding 84% (6–90%)
Fecal calprotectin[b] 43% (8–51%)
C-reactive protein 64% (5–68%)
Inpatient visits 27% (15–42%)
Emergency department visits[b] 51% (13–64%)
Inpatient steroid use 29% (1–30%)
Outpatient steroid use 28% (21–49%)
0 10 20 30 40 50 60 70 80 90 100
Patients (%)

[a]Present or calculated.

[b]One client was excluded due to absence of data availability.

Limitations

- Parameter data may be limited due to challenges in extracting PROs from clinical notes and the incomplete capture of external laboratory tests and/or procedures
- Evidence Council participant views may not be representative of all health systems and specialty pharmacy settings

Conclusions

- **This project highlights a need for standardizing UC disease activity assessments to enable specialty pharmacy practices to engage patients and address existing practice gaps**
- **Evidence-based approaches for UC disease assessment and treat-to-target are available, but are not being followed in clinical practice, limiting outcome analyses**
- **Specialty pharmacies have an important role in collecting PRO and quality of life data that are necessary to inform treatment decisions**
- **There is an opportunity for specialty pharmacies to align on disease-specific parameters to streamline UC patient management, facilitate population health approaches, and ultimately improve patient outcomes**

Electronic poster

https://scientificpubs.congressposter.com/p/hfy1q7dsh9vlbr4m

Abbreviations

IBD, inflammatory bowel disease; **MeSH**, medical subject headings; **PRO**, patient-reported outcome; **PRO2**, patient-reported outcomes 2; **TSOC**, Therapy-Specific Outcomes Coalition; **UC**, ulcerative colitis.

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Disclosure of interests

MR, **MG**, and **SK** have nothing to disclose. **SM** and **JL** are employees and shareholders of Pfizer Inc.

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